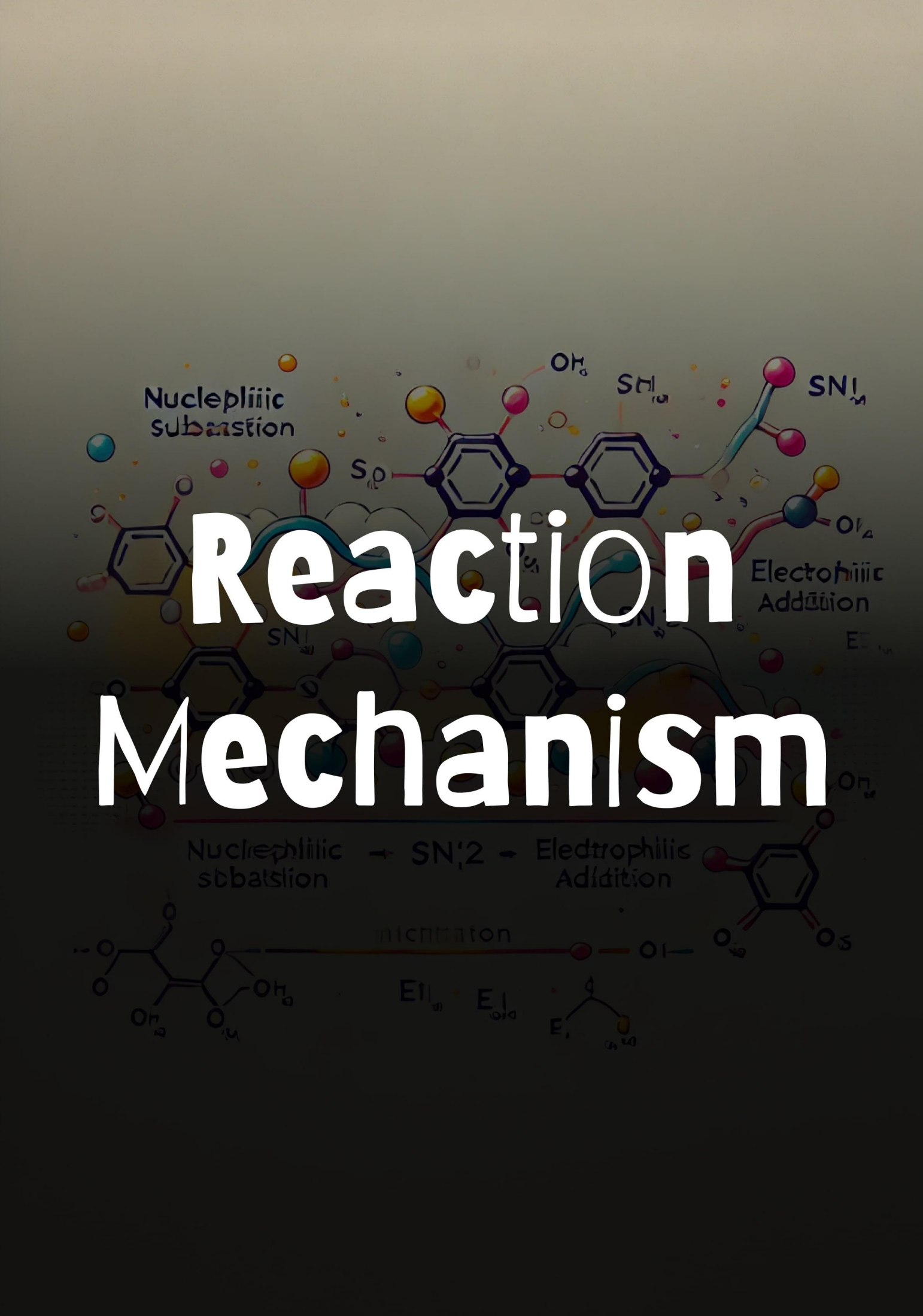

Reaction
Mechanism

MECHANISM

Mechanism describes how the reaction proceeds it describes how the bond breaks and forms. It describe the nature of the intermediates. The major and minor products ; it describes the stereochemistry of the products formed; it describes the solvent effect, it defines everything related to reaction.

On the basis of breaking, forming of bonds ; organic reactions are classified into three groups :-

a. By heterolytic cleavage
b. By homolytic cleavage
c. Concerted mechanism

a. By heterolytic cleavage

$$Z^{\ominus} + R-X \longrightarrow R-Z + X^{\ominus}$$

$$A-B \longrightarrow A^{+} + B^{-}$$

The reaction intermediates are $R^{\oplus}$ or $R^{\ominus}$

(i) $R-X \rightleftharpoons R^{\oplus} + X^{\ominus}$ ⟶ (By heterolysis)

(ii) $R^{\oplus} + Z^{\ominus} \longrightarrow R-Z$

Thus, there must be minimum two stages the slow step is the first step (R.D.S) ⟶ slow step ⟶ Gen^{n} of $R^{\oplus}/R^{\ominus}$. $1 \longrightarrow 1+1$, thus, the entropy ↑, it is called the entropy.

All heterolysis reac^{n} will show +ve entropy of activation.

b. By homolytic cleavage

$$R-H + X_2 \xrightarrow{h\nu} R-X + H-X$$

Initiation ← i $X_2 \xrightarrow{h\nu} 2X\cdot$

Propagation ← ii $R-H + X\cdot \longrightarrow R\cdot + HX$

iii $R\cdot + X-X \longrightarrow R-X + X\cdot$

Termination ← iv $R\cdot + X\cdot \longrightarrow R-X$

v $R\cdot + R \longrightarrow R-R$

vi $X\cdot + X \longrightarrow X-X$

All radicals are Paramagnetic

First step is always generation of radicals. It is called initiation. It is always endothermic. Radical cond^n ↑.

Propagation is a chain reaction as radical reacting is regeneration, Thus, radical reaction stand still. May be exo/endo.

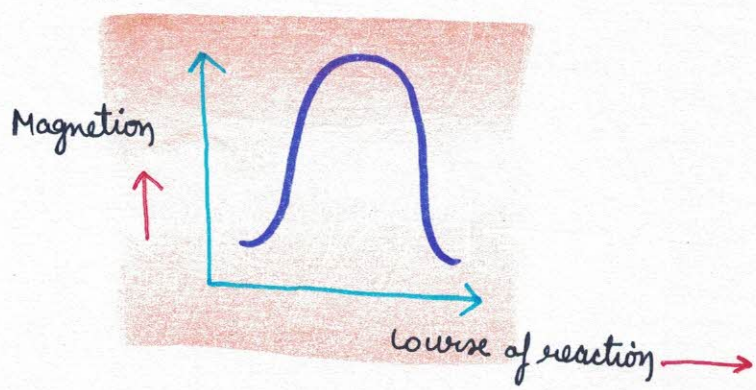
Termination is always exothermic.

If propagation is exothermic, it is irreversible chain rxn.

If propagation is endothermic, it is reversible chain rxn.

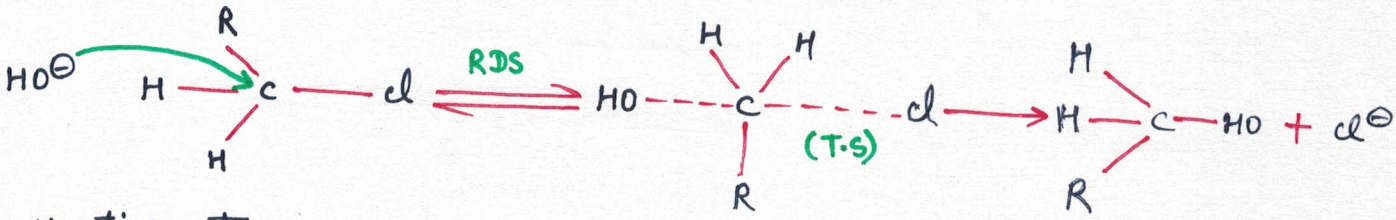
Thus, homolysis is a multi-stage reaction.

Thus, during reaction (Initiation, propagation) → The reaction mix. show Paramagnetism due to radicals Termination → No paramagnetism, thus, no magnetic activity.



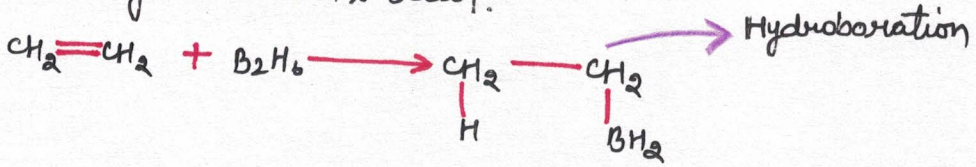
CONCERTED MECHANISM

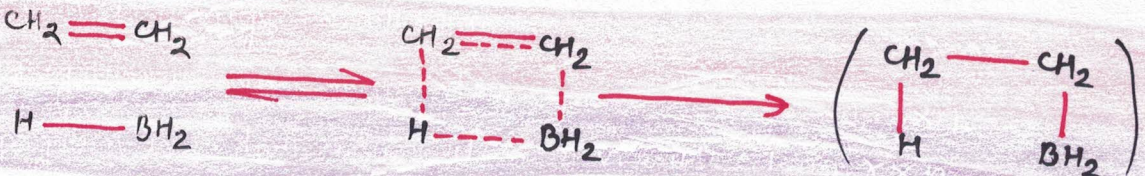
Bond forms and bond breaks simultaneously in a sequence, it is a single step. Pentavalence carbon is intermediate (T.S)



Negative entropy as the T.S. is more crowded.

Rearrangement doesn't occur.





PERICYCLIC MECHANISM

when the cyclic T.S. is an intermediate then it is pericyclic. All pericyclic → concerted. But all concerted are not pericyclic.

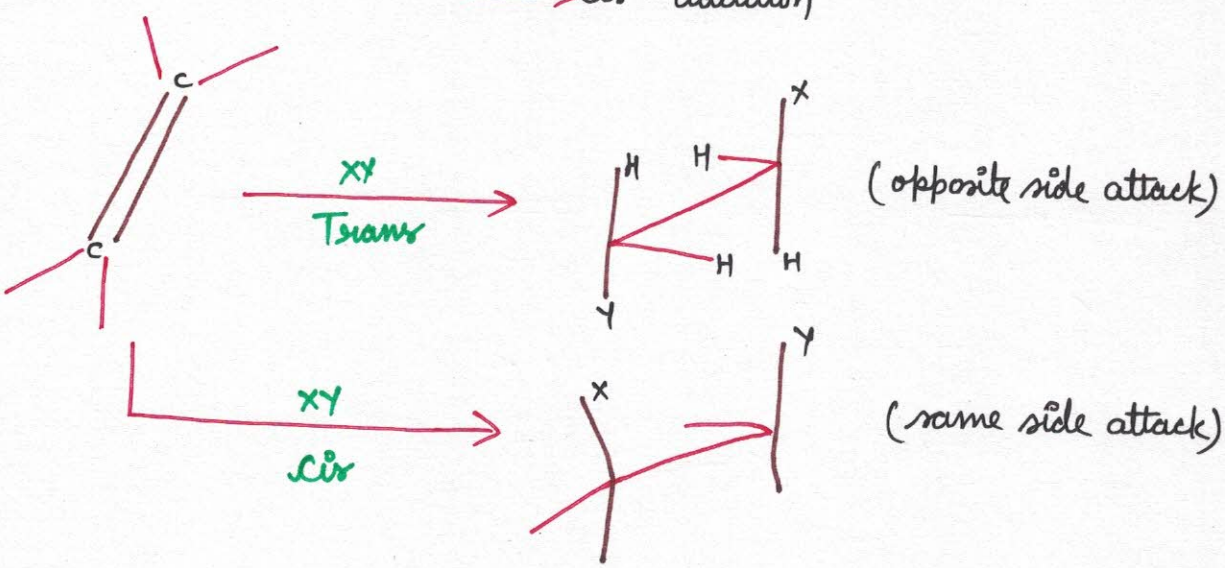
TYPES OF REACTIONS

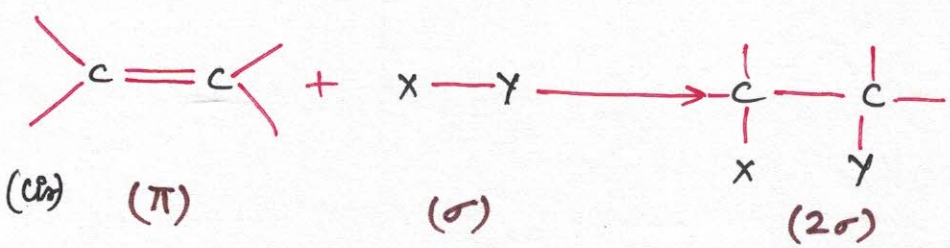
Addition, elimination, substitution, Reduction, oxidation, Rearrangement.

(i) ADDITION REACTION

1 + 1 = 1

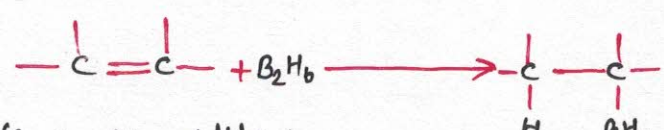
Addition Reaction → Trans - addition
Addition Reaction → Cis - addition





All addition reaction are exothermic ($\Delta H < 0$)
Entropy — temp. must be lowered.
if $\Delta G < 0$, then reaction is favourable.
if $\Delta G = 0$, then reversible (At equilibrium)
if $\Delta G > 0$, then no reaction occurs.

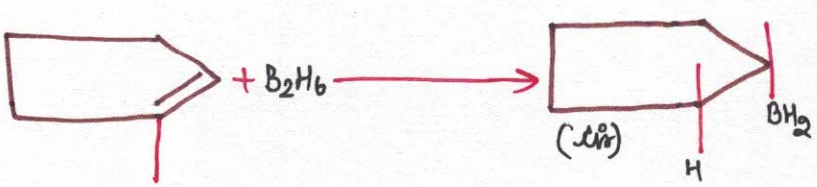
Cis addition



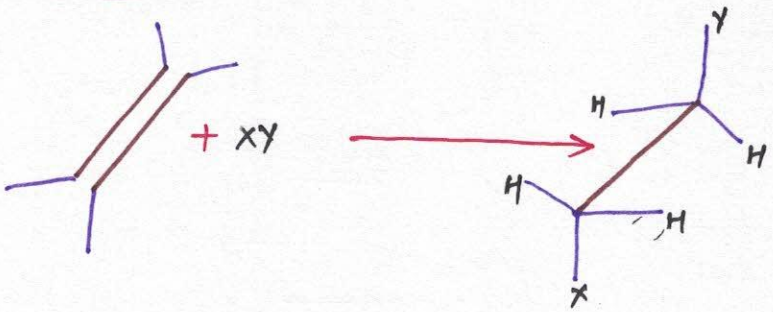
(Hydroboration cis addition)

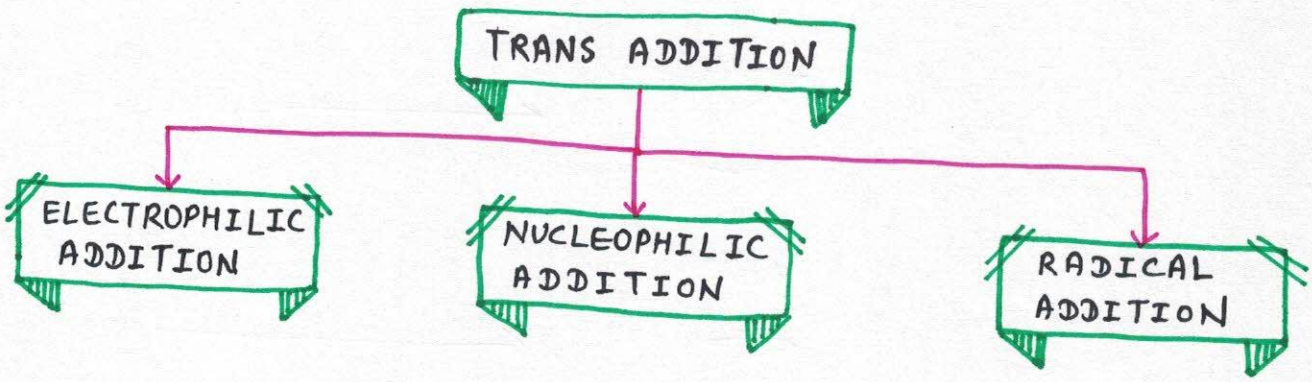
(Pericyclic addition)

All cyclic - addition are cis - addition



Trans-addition

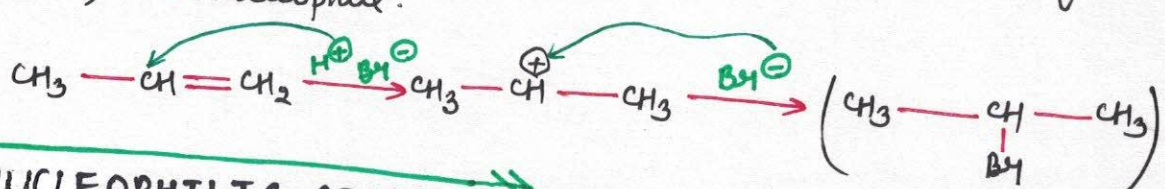




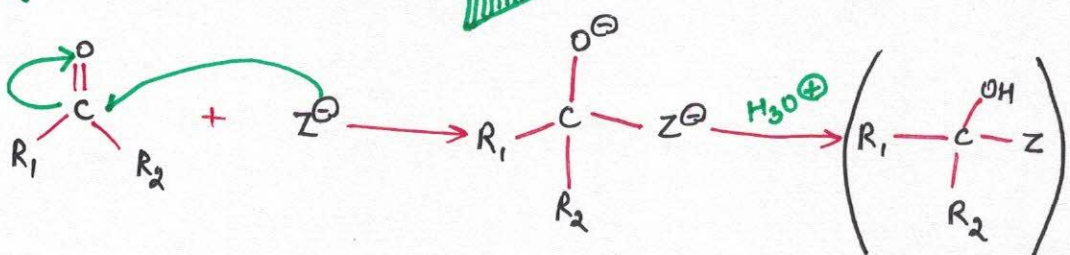
a. ELECTROPHILIC ADDITION



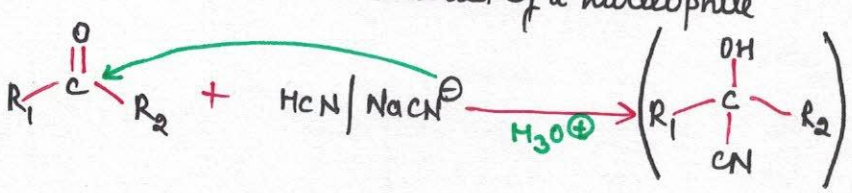
Initiated by electrophile, thus, electrophilic. It can be termed as acid-base rxn. Alkenes only undergo the electrophilic addition as they have lp (basic) → Nucleophile.

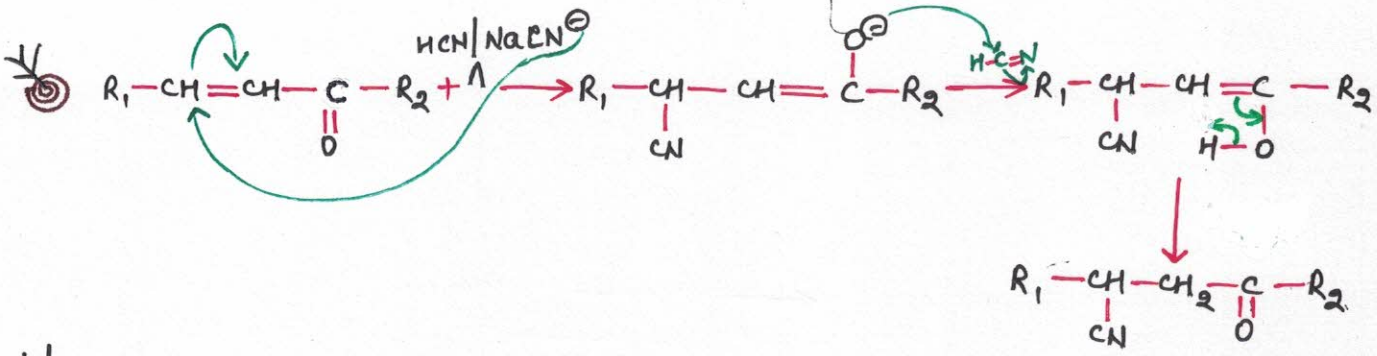


b. NUCLEOPHILIC ADDITION

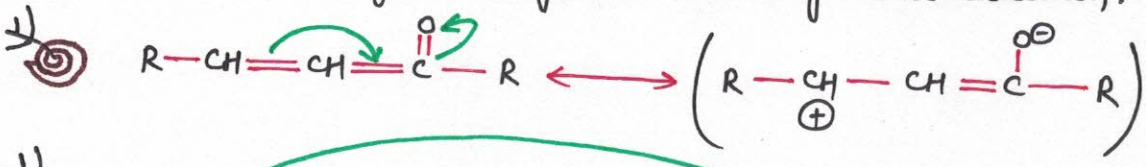


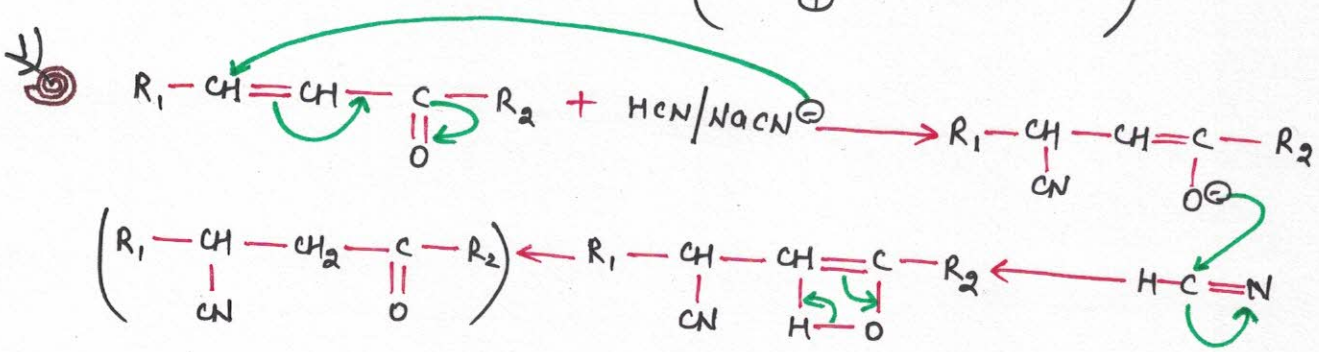
Initiated by a nucleophile



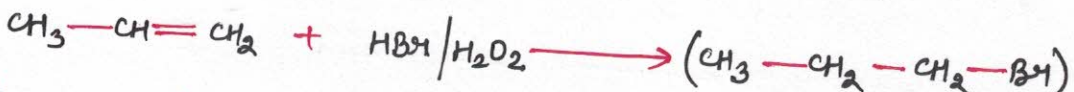


- Esters, carbonyls, anhydrides undergo nuc. addition.

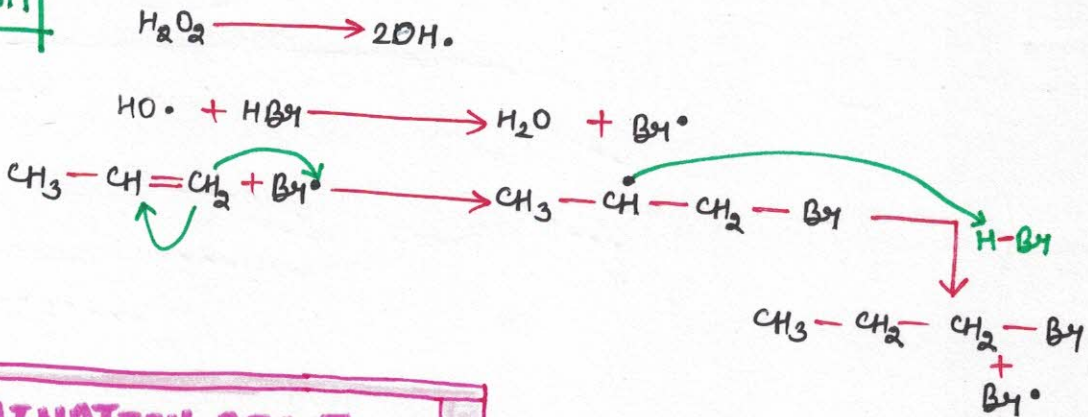




c RADICAL ADDITION



MECHANISM



(ii) ELIMINATION REACTIONS

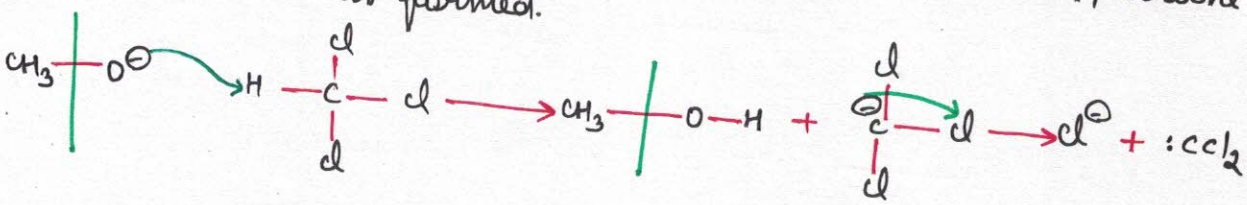


ELIMINATION

- 1,1-elimination (α-elimination)
- 1,2-elimination (β-elimination)

(a) α-elimination

- Two groups are removed from same carbon, carbene is formed.



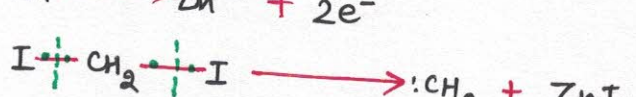


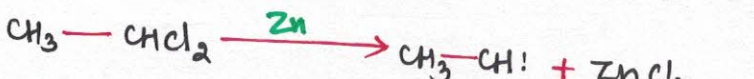
(chlorofluorous carbene)

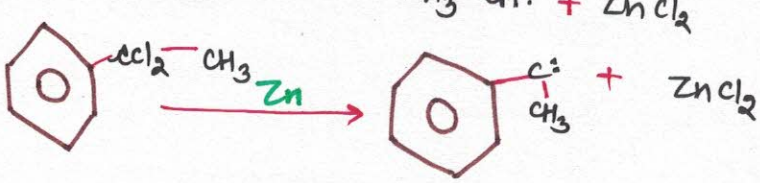


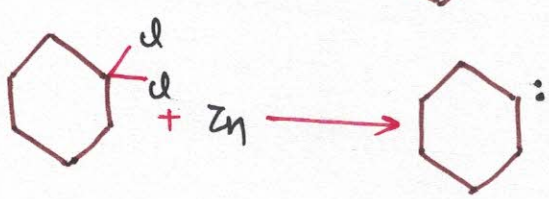
MECHANISM







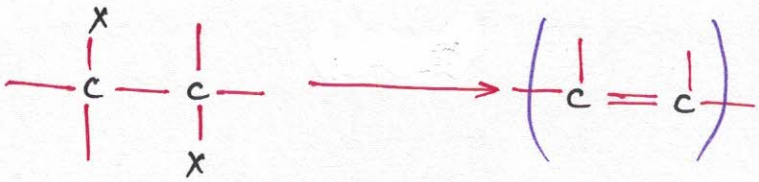


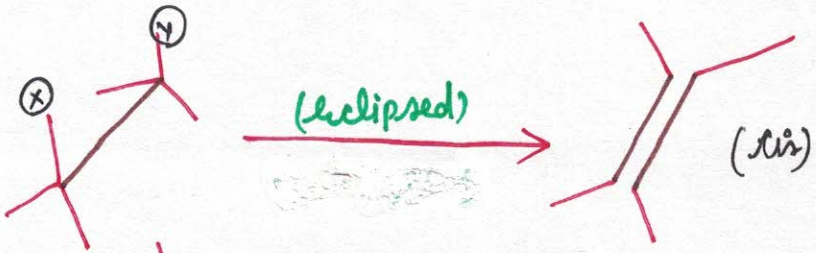


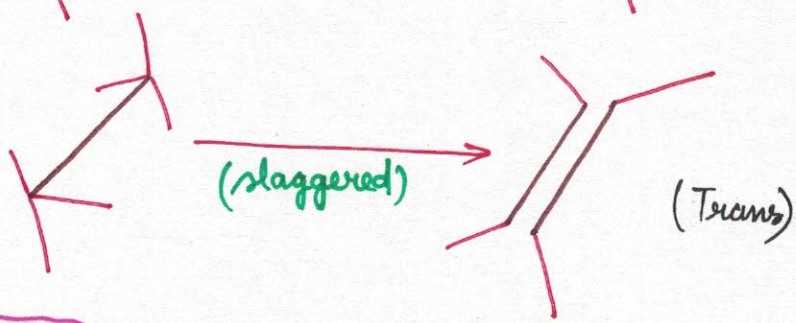
(b) 1,2-elimination (β)

- → Cis elimination
- → Trans elimination
  - → E_1
  - → E_2
  - → E_{1cB}

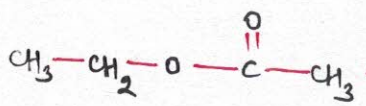
Introduction of a π bond by removal of 2 adjacent groups.







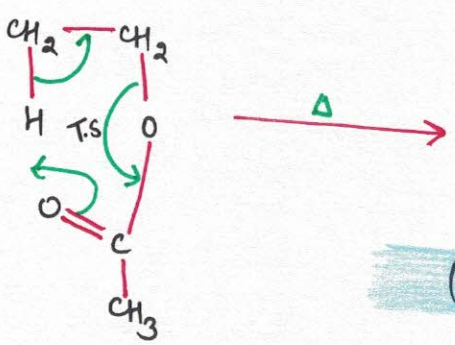
Cis elimination



(ethyl ethanoate)

$CH_2 = CH_2 + CH_3COOH$

(Pyrolysis — Heating without O_2)



$CH_2 = CH_2 + CH_3COOH$

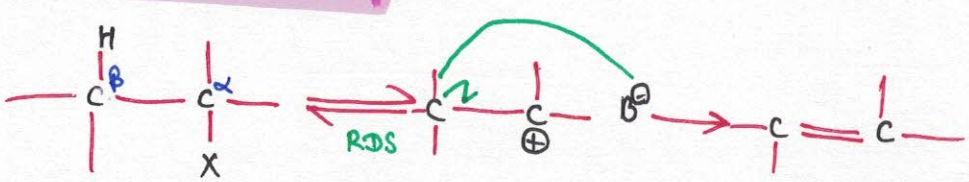
(bond breaks and bond forms)

(Pericyclic) → Cis elimination

Pyrolysis of esters is a pericyclic (cis elimination)

Trans elimination

(a) E_1 elimination



Rate $\propto$ [RX] Rate depends on how fast the ion pair is formed. Rate doesn't

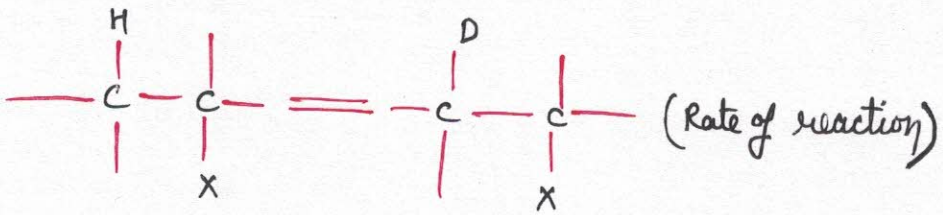
depend on the conc^{n} of base. Also known as elimination unimolecular.

If $C^{\oplus}$ is an intermediate, then it is always E_1.

Associated with rearrangement, then it is always effect. If the rate of reaction is proportional to leaving group ability. the reaction is said to be showing to be elemental effect.

In all E_1 reactions $R-I > R-Br > R-Cl > R-F$

It doesn't show β-hydrogen isotopic effect.



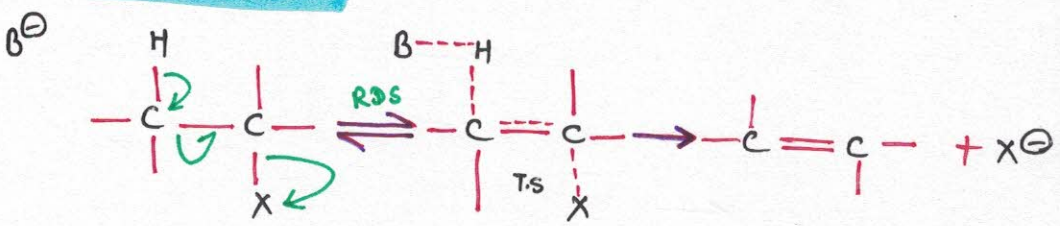
(Rate of reaction)

$CH_3-CH_2-Cl = CD_3-CH_2-Cl$ (Rate of reaction)

It is a trans- elimination. Doesn't show solvent isotopic effect.

shows +ve entropy of activation (7 bonds → 6 bonds)

(b.) E_2 elimination

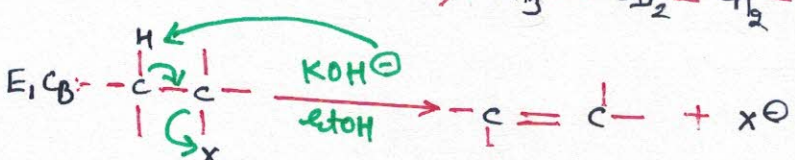


Rate ∝ [RX][B⁻]. elimination bimolecular [E_2]. concerted mechanism T.S is an intermediate. No rearrangement. show elemental effect. shows β-hydrogen isotopic effect. (H D), Doesn't show solvent isotopic effect. Trans. elimination.

shows -ve entropy of activation.

$CH_3-CH_2-Cl = CH_3-CD_2-Cl$ (Not α-H but β-H)

$CD_3-CH_2-CD_2-Cl > CH_3-CD_2-CH_2-Cl$ (Rate of reaction)



Rate depends on how fast the carboanion is formed.

(c) E, cB - Elimination

Elimination unimolecular of c.B. no rearrangement occurs. It shows β-hydrogen isotopic effect Doesn't show elemental effect. shows the solvent isotopic effect.

The c-x bond breaks after slow step but not during the slow step. Thus, no elemental effect.

R-F > R-Cl > R-Br > R-I

SOLVENT EFFECT (ISOTOPIC)

If there's an exchange of an isotopic from the solvent to reactant, it is said to be showing solvent isotopic effect. All those organic rxn. where $c^{\ominus}$ is an intermediate will show the solvent isotopic effect.

If during the course of rxn; the mol. weight rises by (M+1 / M+2 / M+3), then it shows solvent isotopic effect.

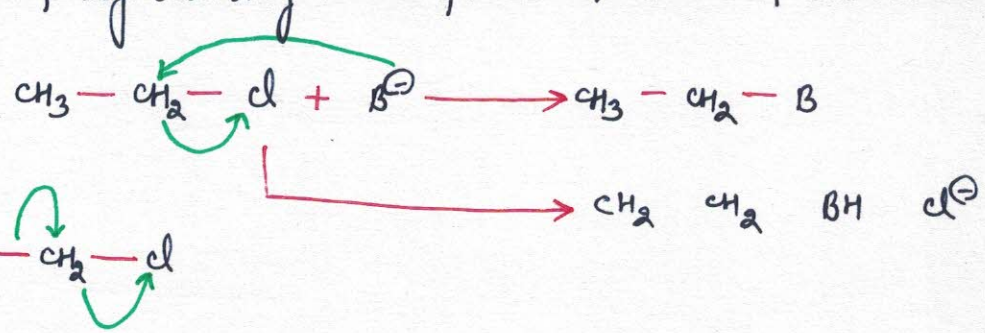
Elimination reaction are endothermic. ΔS 0. entropy will ↑. elimination reaction must be done at the higher temp, under vigorous condition.

(iii) Substitution Reaction

1+1 ⟶ 1+1

Substitution only occurs if reaction is exothermic. It occurs at any temp

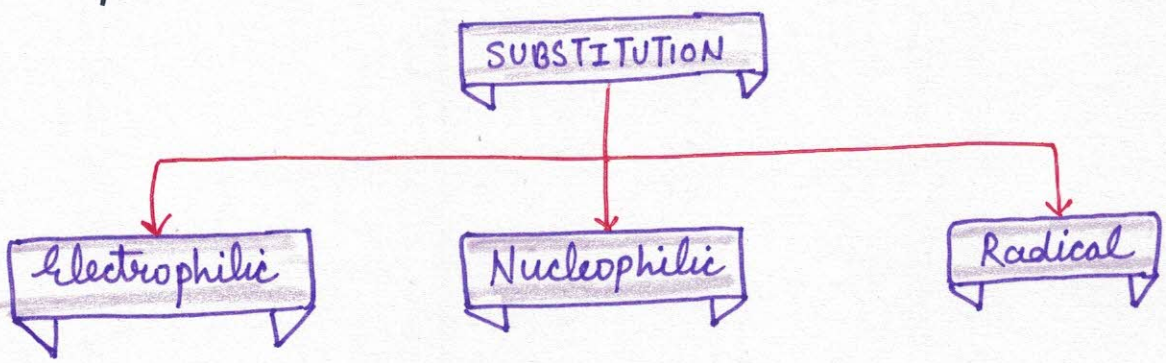
ΔS = 0



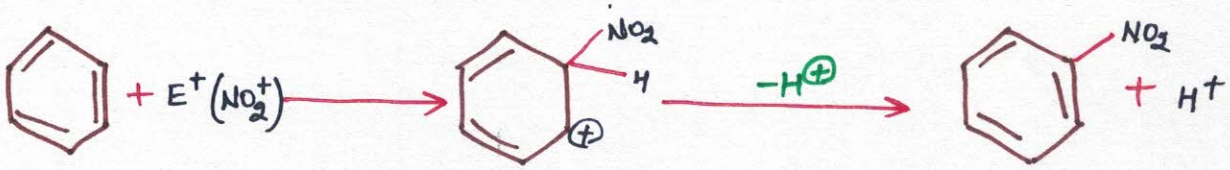
Elimination and substitution go hand-in-hand.

For only substitution to occur use "weak base" like aq. NaOH and lower temp.

For only elimination to occur, use "strong base" like alc. KOH and higher temp.



(a) Electrophilic



All aromatic hydrocarbons undergo electrophilic substitutions.

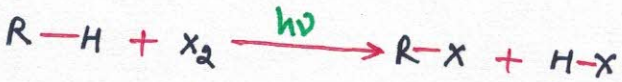
(b) Nucleophilic



S_N1 S_N2 S_Ni by N.G.P. S_N2, S_Ni

Alkyl hallides undergo nucleophilic substitution (Acids, Alcohols, etc).

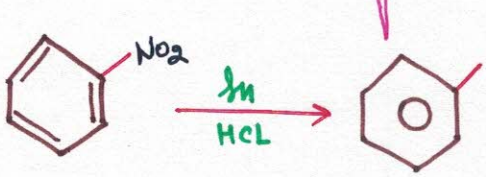
(c) Radical substitution



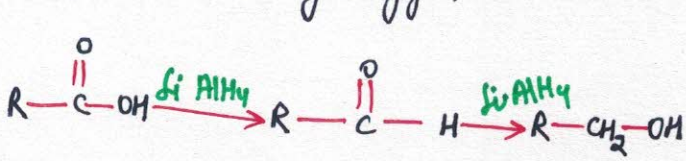
It proceeds through the radical mechanism. Hallogenation of alkene.

iv REDUCTION

Addition of hydrogen, removal of oxygen.

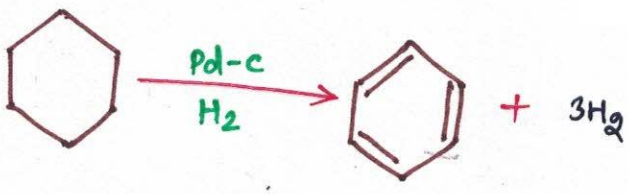


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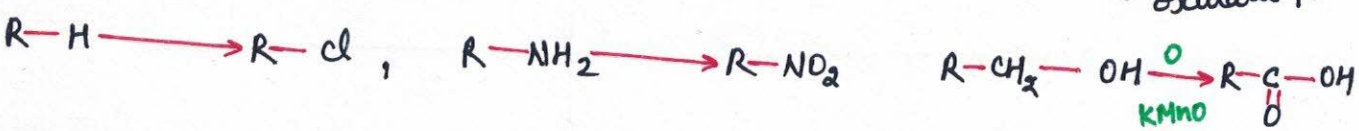




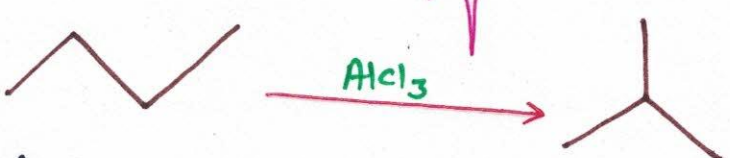
(V) OXIDATION



Hallogenⁿ is oxidation.

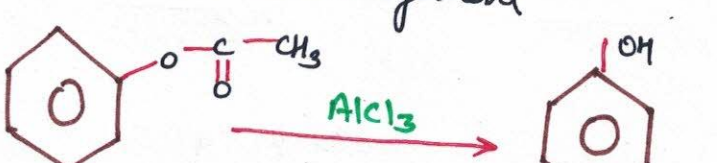


(VI) REARRANGEMENT

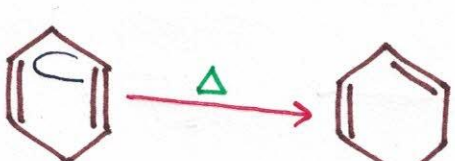


(Isomerisation)

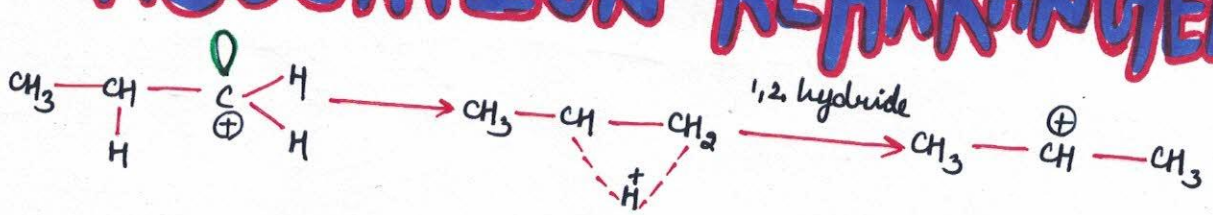
Pinacol-Pinacolone rearrangement



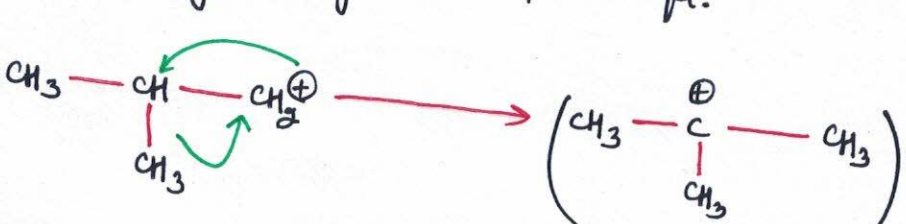
(Fries)

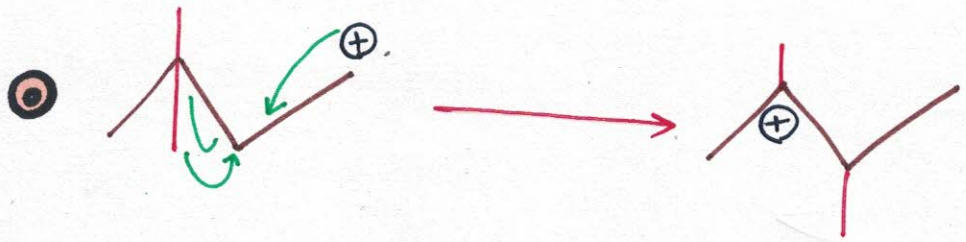


CARBOCATION REARRANGEMENT

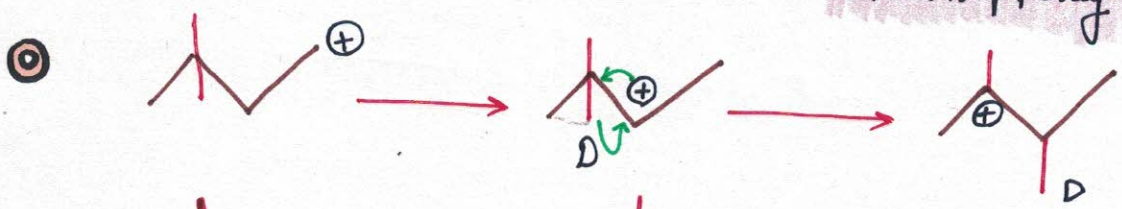


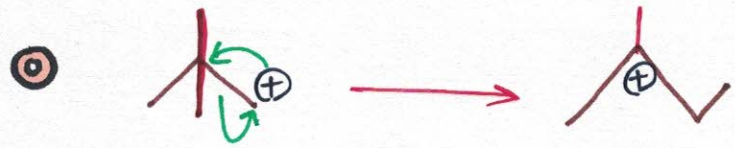
⊕ only undergo the -1,2- shift.

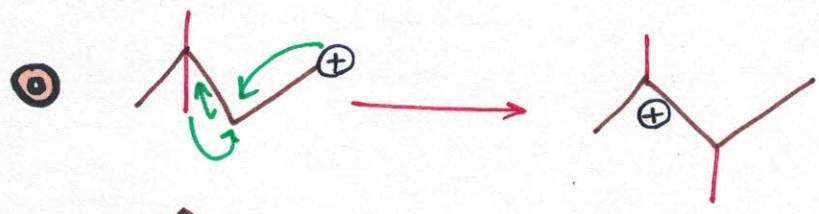


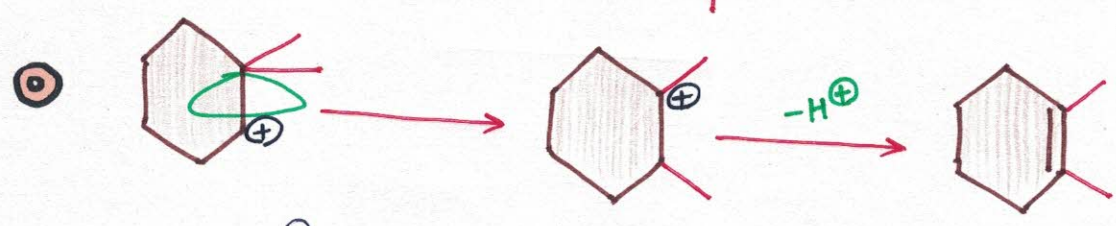


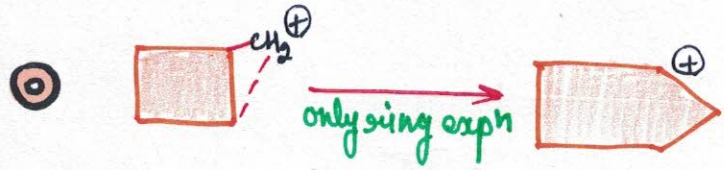
In one step, only shift happens.



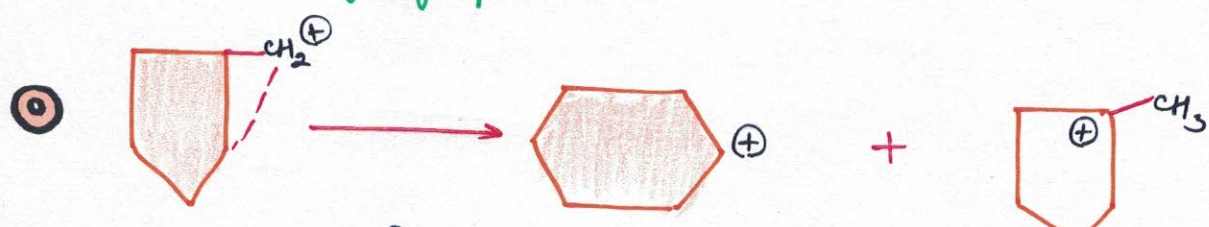


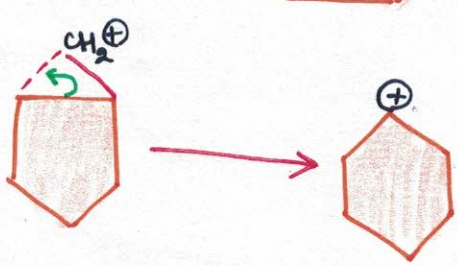




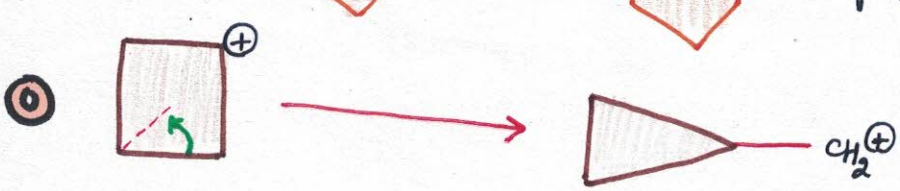


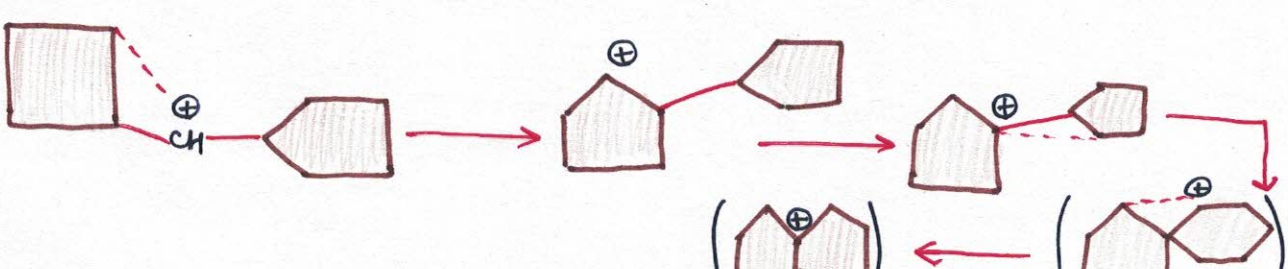
(cyclobutyl methyl Ring expansion)

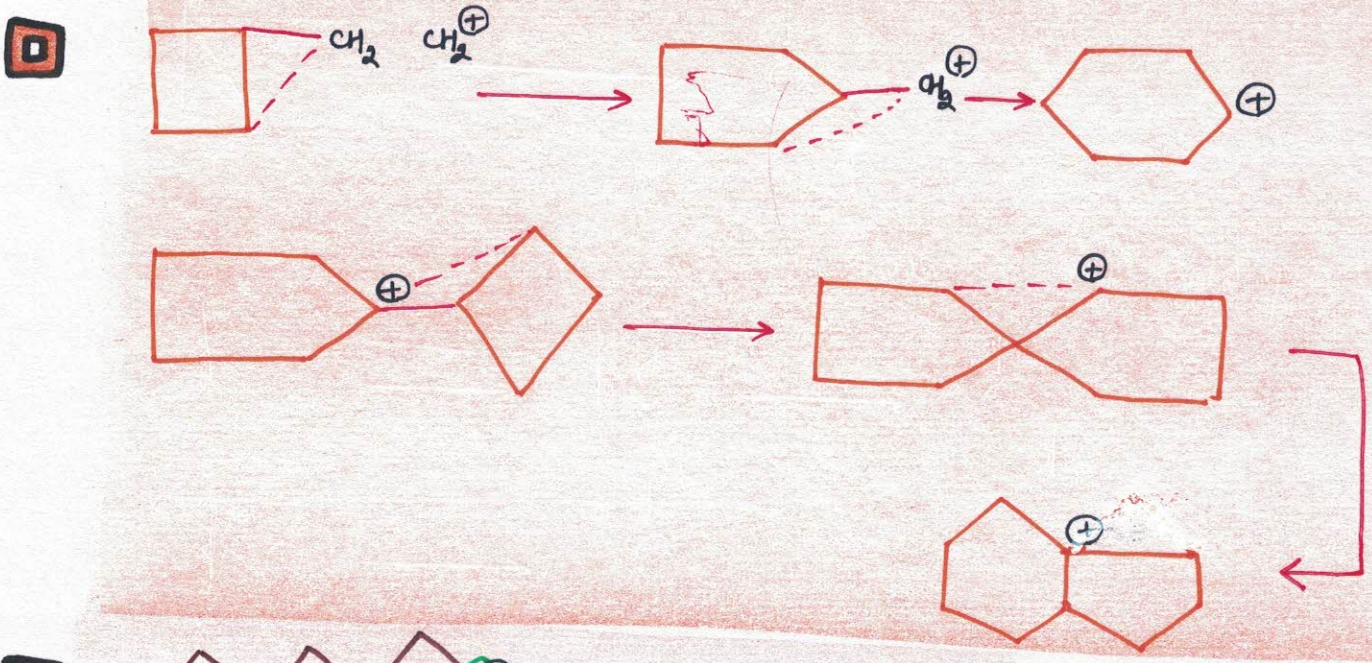


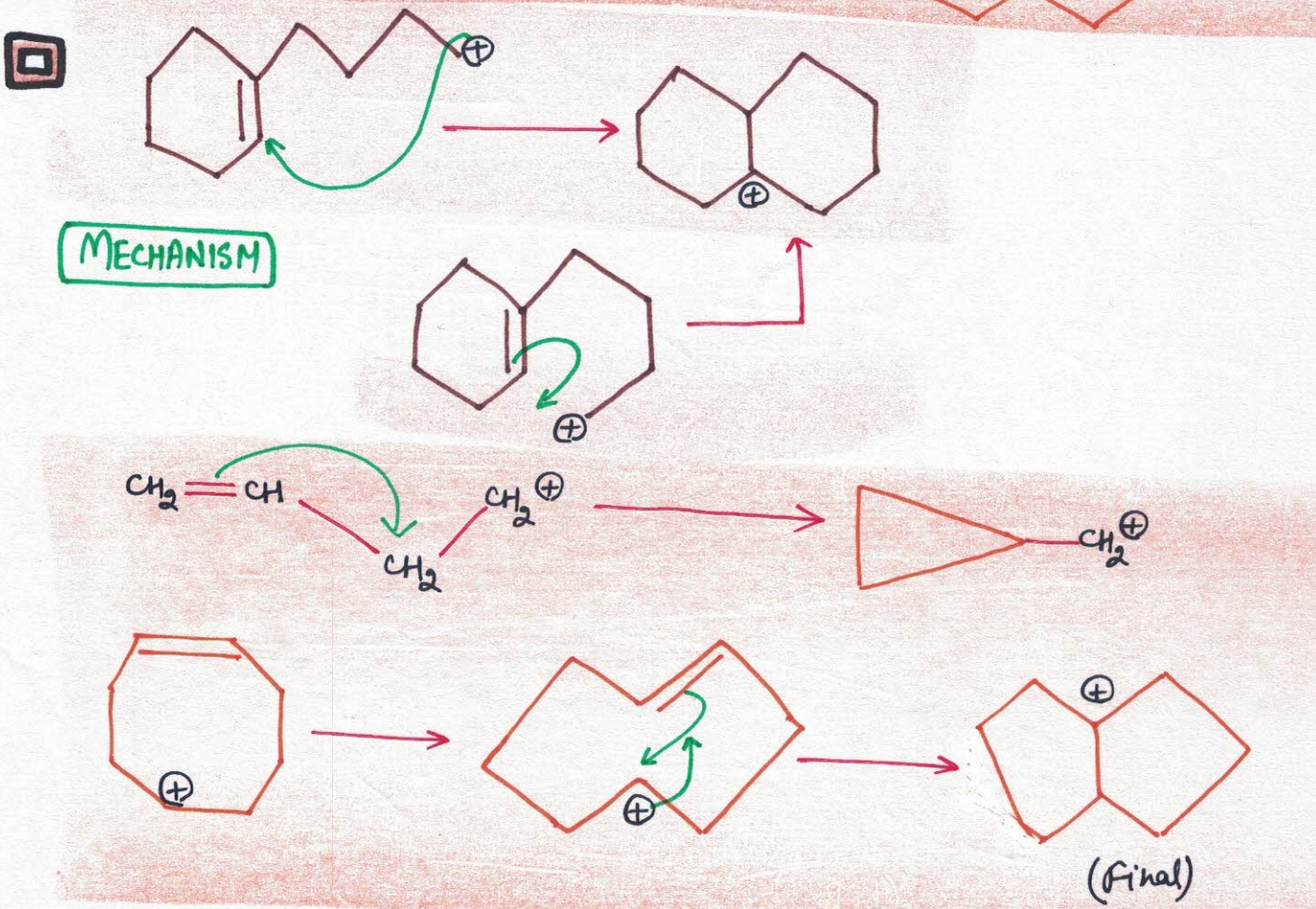


(major)

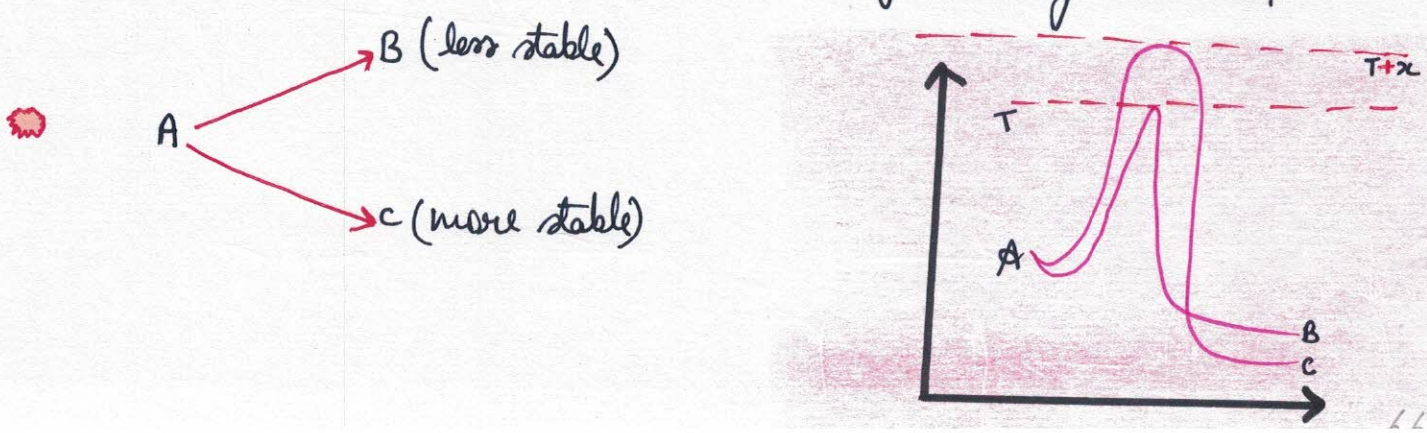




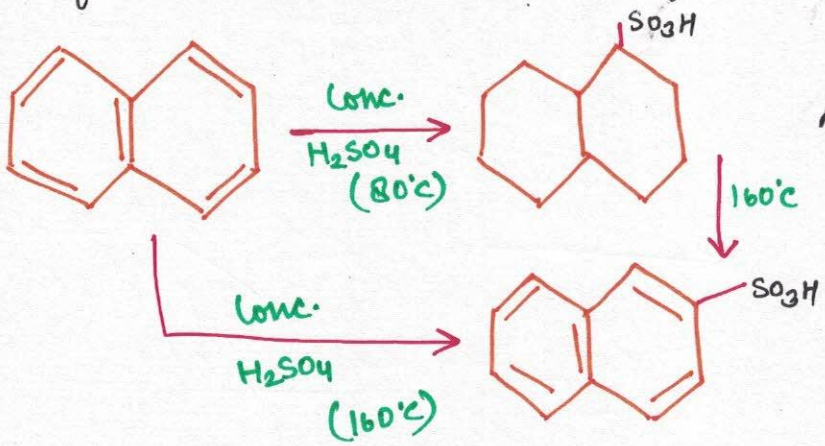


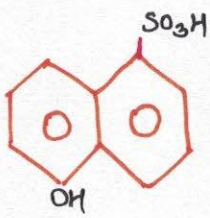


- Kinetically controlled reactions and Thermodynamically controlled.



Energy of activation for A → B is less than for A → C

- At temp. T, only A B and A C won't occur.
  At temp T x, A B and A C; but the quantity of B formed will be much more
- If the yields are proportional to rate of reaction, the r n. is said to be Kinetically controlled.
- B A B, thus, A B will be reversible as the energy of activation is low but for A C, the energy of activation is high; thus A C not reversible.
- At lower temp, the prod. will be B.
- If the yields are proportional to stability of prod, the reaction is said to be thermodynamically controlled.
- Kinetically controlled prod. is easier to form.
- 

- Kinetically controlled prod. changes to thermodynamically controlled product on heating.
- Due to peri-hydrogen repulsion, 

is less stable
- 